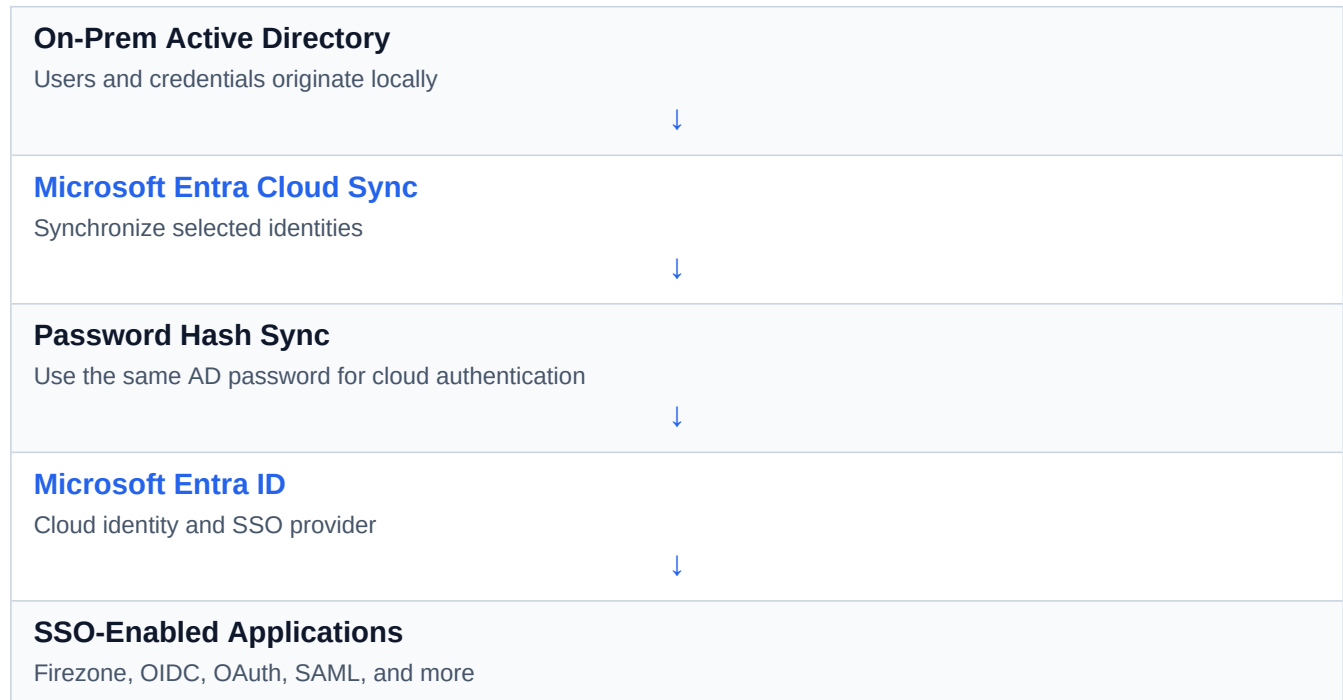


Enterprise-Style SSO at Home

Using On-Premises Active Directory + Microsoft Entra ID Free



A practical, step-by-step guide for building a reusable hybrid identity foundation.

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What You Are Building

This guide walks through a small-scale hybrid identity design that keeps Windows sign-in on local Active Directory while extending those same identities into Microsoft Entra ID for cloud authentication and SSO. The design is intentionally simple, controlled, and reusable for a home lab.

Key outcome

A user can sign in to a supported application using an identity that originates in local Active Directory, synchronizes to Entra, and authenticates through Entra SSO - without replacing normal domain authentication on Windows.

Before You Start

- A working on-premises Active Directory domain and at least one test user.
- A Microsoft Entra tenant using the Free plan.
- A public domain you control and can modify in DNS.
- A domain-joined Windows Server that can run the Microsoft Entra provisioning agent.
- Internet access from the provisioning-agent server.
- An application that supports Microsoft Entra, OpenID Connect (OIDC), OAuth 2.0, or SAML.

Recommended approach

Start with one test user and a small synchronization scope. Prove the flow end to end before adding more users or applications.

Implementation at a Glance

Step	Area	Goal
1	Entra domain	Verify your public domain
2	Active Directory	Confirm user UPNs align
3	Entra admin	Create a cloud-only sync admin
4	Provisioning agent	Install and register Cloud Sync agent
5	AD scope	Create a security group for synchronized users
6	Cloud Sync	Enable password hash sync and configure scope
7	Testing	Provision one user on demand
8	Deployment	Review and enable the configuration
9	Authentication	Verify AD password works in Entra
10	SSO application	Connect Firezone or another supported app

1 Add and Verify Your Public Domain in Entra

Your public domain must be verified in Microsoft Entra ID if you want synchronized users to sign in with a familiar UPN such as user@example.com.

1. Open Microsoft Entra admin center.
2. Go to Entra ID > Domain names > Add custom domain.
3. Enter the public domain you control.
4. Create the TXT (or MX) record Microsoft provides at your DNS provider.
5. Return to Entra and verify the domain.

Why this matters

If the on-premises UPN suffix is a verified Entra domain, the synchronized identity can keep the same user@example.com sign-in name end to end.

2 Check On-Premises Active Directory UPNs

In Active Directory Users and Computers, open a test user and check Properties > Account. The User logon name should ideally match the verified public domain.

```
user@example.com
```

If your AD forest uses a different or non-routable suffix, add the public domain as an alternate UPN suffix rather than renaming the Active Directory domain.

3 Create a Cloud-Only Entra Admin

Create a dedicated Entra account for Cloud Sync administration instead of relying only on the identity you are about to synchronize.

```
cloudsyncadmin@example.com
```

- Create the user directly in Entra.
- Assign the Hybrid Identity Administrator role.
- Complete the temporary password change.
- Complete MFA / security-info registration before using the account in the provisioning-agent installer.

Separate the credential sets

The Entra admin authenticates the cloud side. Your on-prem domain admin is used later to configure the local AD side and gMSA.

4

Install the Microsoft Entra Provisioning Agent

In Entra, go to Entra Connect > Cloud Sync > Agents and download the on-premises provisioning agent. Install it on a domain-joined Windows Server.

- Connect Microsoft Entra ID: sign in with the cloud-only Hybrid Identity Administrator.
- Configure Service Account: choose Create gMSA unless you already manage a custom gMSA.
- Provide on-premises domain administrator credentials when prompted.
- Connect Active Directory: confirm your AD domain is discovered and added.
- Confirm the configuration and finish the wizard.

Validation checkpoint

Return to Entra Connect > Cloud Sync > Agents. Do not continue until the agent shows Active.

5

Create a Controlled Synchronization Scope

For a lab or pilot, use a dedicated AD security group so you can explicitly control which users synchronize to Entra.

```
Entra-Sync-Users
```

- Group scope: Global
- Group type: Security
- Add test users directly as members.

Retrieve the group Distinguished Name (DN) with PowerShell:

```
Get-ADGroup "Entra-Sync-Users" | Select-Object DistinguishedName
```

A result might look like:

```
CN=Entra-Sync-Users,CN=Users,DC=example,DC=com
```

Nested groups

For a simple pilot, add users directly to the scope group. Do not depend on nested membership being evaluated the way you expect.

6 Create the Cloud Sync Configuration

In Entra, go to Entra Connect > Cloud Sync > New configuration, choose AD to Microsoft Entra ID, select the AD domain, and enable Password Hash Sync.

6. Create the configuration.
7. Open Scoping filters.
8. Choose Selected security groups.
9. Paste the DN for Entra-Sync-Users.
10. Add the group and save the scope.

Password Hash Sync

This allows a synchronized Entra user to authenticate to cloud services with the same password used in local Active Directory.

7 Test One User with Provision on Demand

Before enabling recurring synchronization, test a single account. Provision on demand expects the AD object Distinguished Name rather than the UPN.

```
Get-ADUser username | Select-Object DistinguishedName
```

Example:

```
CN=John Smith,CN=Users,DC=example,DC=com
```

11. Open Provision on demand in the Cloud Sync configuration.
12. Paste the user DN.
13. Run the test.
14. Confirm the export step reports that the user was created in Microsoft Entra ID.

Success criteria

In Entra ID > Users, the new identity should appear with On-premises sync = Yes and the expected UPN.

8 Review and Enable the Configuration

After the single-user test succeeds, return to the Cloud Sync configuration and choose Review and enable. The scoped users will begin synchronizing automatically.

Local authentication

Windows sign-in continues to use your on-premises Active Directory exactly as before.

Cloud authentication

Entra receives the synchronized identity and password hash for supported cloud sign-in and SSO.

Verify Password Hash Synchronization

Sign in to a Microsoft authentication page with the synchronized user and the normal on-prem AD password. A short delay can occur after the user object is first created before password authentication succeeds.

If you see invalid username or password

First give Password Hash Sync several minutes to catch up. If the identity exists but the password still fails, check Cloud Sync and provisioning logs before changing the architecture.

Connect an SSO Application

Once the synchronized account can authenticate to Entra, connect an application. Some services have a built-in Microsoft Entra provider; others use generic OIDC, OAuth 2.0, or SAML settings.

- Tenant ID
- Client ID
- Client secret (when required)
- Redirect URI
- Issuer / authority URL
- Claims and scopes

Keep a fallback during testing

If the application already has a working login method, leave it enabled until Entra SSO is fully tested. This avoids locking yourself out while experimenting.

Example: Firezone

Firezone can use Microsoft Entra as an identity provider while its existing email OTP authentication remains available. This makes it a good first SSO test because you can validate Entra without disrupting users who sign in with different email addresses.

Matching identity

If the Firezone user email matches the Entra UPN, the identity can be associated with that existing user for SSO.

Non-matching identity

Users with different Firezone email addresses can continue using the existing Firezone login method.

The final flow becomes:

```
On-Prem AD credentials -> Entra authentication -> Firezone SSO
```

The first-time Firezone Entra provider verification may require a tenant-wide Global Administrator or Privileged Role Administrator to grant consent. After the provider is configured, normal synchronized users do not need that elevated role just to sign in.

What This Setup Changes - and What It Does Not

What it adds	What stays the same
Cloud identity for selected AD users	Normal Windows domain login
Password Hash Sync for Entra auth	On-prem AD remains authoritative for local sign-in
Reusable SSO foundation	Existing GPOs and local AD permissions
Support for Entra/OIDC/OAuth/SAML apps	Non-SSO app authentication unless you change it

Entra Free: Practical Boundaries

The Free plan is sufficient for the core lab described here: user and group management, on-premises directory synchronization, password-hash-based cloud authentication, and SSO for supported applications. More advanced identity governance and access-control features can require paid Entra licensing.

Design around the Free tier

Treat this as an identity and SSO lab. Do not assume every enterprise feature shown elsewhere in Entra - especially advanced Conditional Access and risk-based controls - is included in the Free plan.

Troubleshooting Checklist

Agent not visible or inactive

Confirm the provisioning agent service is running, the server has outbound internet access, and the agent is registered to the correct tenant.

User not found during Provision on demand

Use the user Distinguished Name, not the email/UPN. Also confirm the user is directly in the scoped security group.

User appears in Entra but password fails

Wait for Password Hash Sync, then review Cloud Sync / sign-in logs. A successful object create does not prove the password hash is immediately usable.

SSO reaches Entra but asks for admin approval

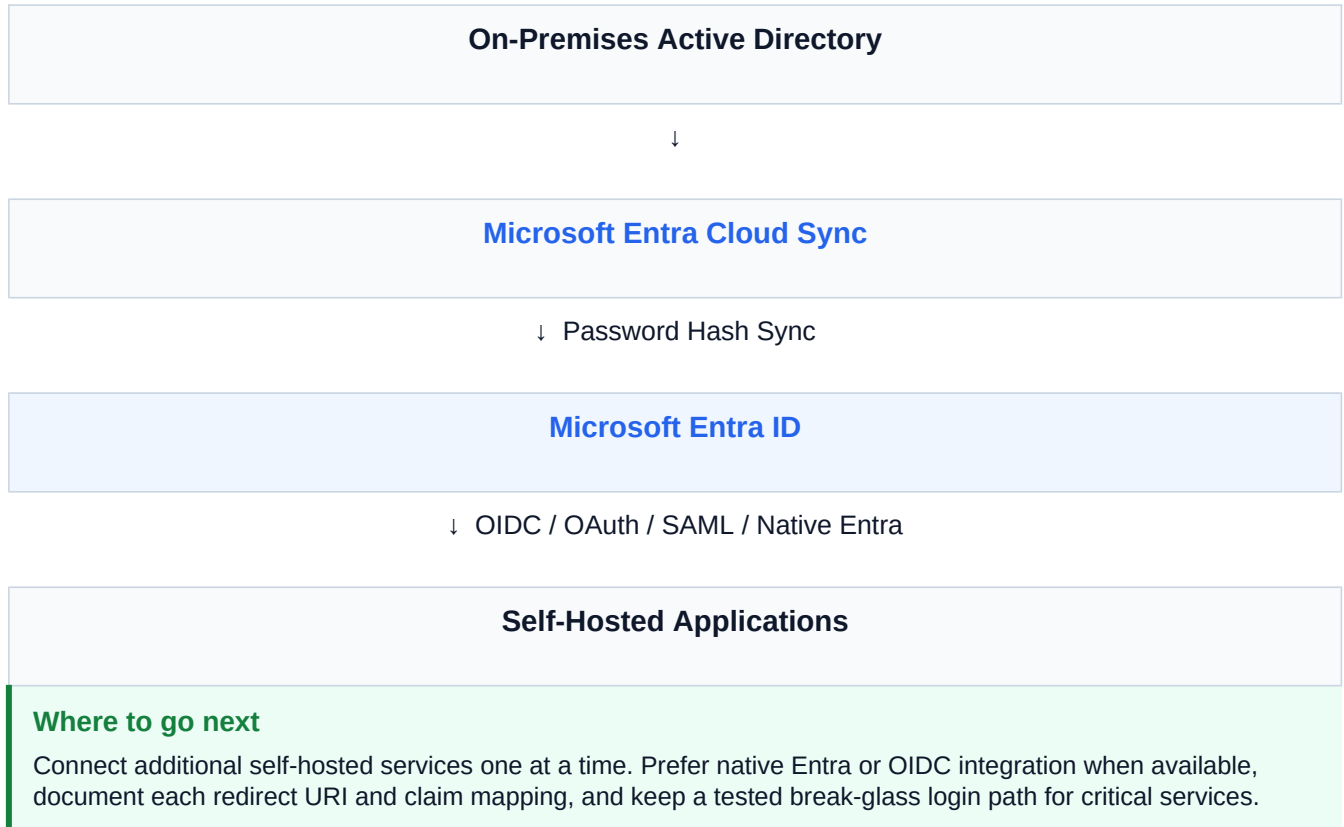
The application is requesting tenant consent. Complete the one-time verification with an appropriately privileged Entra administrator.

SSO user does not match the app account

Check which Entra claim the application uses for email/identity matching and confirm it matches the existing application user.

Final Architecture

Once complete, the home lab has a reusable hybrid identity path that can serve multiple self-hosted applications.



Reference Links

Microsoft Entra ID Free: <https://learn.microsoft.com/azure/cost-management-billing/manage/microsoft-entra-id-free>

Cloud Sync prerequisites: <https://learn.microsoft.com/en-us/entra/identity/hybrid/cloud-sync/how-to-prerequisites>

Configure Microsoft Entra Cloud Sync: <https://learn.microsoft.com/en-us/entra/identity/hybrid/cloud-sync/how-to-configure>

Provision on demand: <https://learn.microsoft.com/en-us/entra/identity/hybrid/on-demand-provision>

Manage custom domain names: <https://learn.microsoft.com/en-us/entra/identity/users/domains-manage>

Firezone Microsoft Entra authentication: <https://www.firezone.dev/kb/authenticate/entra>